SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Assessment Tool Weightage : 20%

QUESTIONS: **Total Marks: 50**

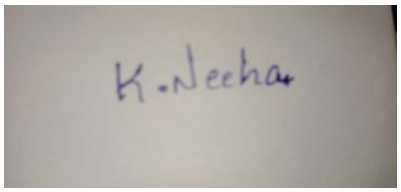
Annexure A

Scenario-Based

Code of Conduct:

I K.Neeha(192425245)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A
Scenario-based

1. Secure Data Transmission using Block Cipher

A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

A. Explain why this issue occurs.
B. Suggest a more secure block cipher mode and justify your choice based on the scenario.

2. Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

A. Identify the most suitable algorithm for this scenario.
B. Justify your choice based on security strength and efficiency.

3. Key Exchange in an Unsecured Network

Two users want to establish a secure communication channel over a public network using Diffie-Hellman key exchange. However, the network is vulnerable to attacks.

A. Explain how the key exchange works in this scenario.
B. Identify a possible security threat and suggest a solution to mitigate it.

4. Public Key Cryptography for Email Security

An organization plans to implement RSA for secure email communication.

A. Describe how RSA can be used for encryption and decryption in this scenario.
B. Explain how key distribution and management play a role in maintaining security.

5. Cryptography for IoT Devices

A smart home system uses low-power IoT devices that need secure communication. The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

A. Analyze the constraints of IoT devices.
B. Recommend a suitable cryptographic approach and justify your answer.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. Introduction

Public key cryptography is a security technique that uses two mathematically related keys: a public key and a private key. The public key can be shared with others, whereas the private key must be kept secret by its owner.

RSA (Rivest–Shamir–Adleman) is a widely known public key cryptographic algorithm used for secure communication, digital signatures, and key establishment. In email security, RSA can help protect confidential messages and verify the identity of the sender.

2. Working of RSA Encryption and Decryption

RSA uses a pair of keys to protect information during transmission from one person to another.

Sender (Alice)

Writes the original email message

Encryption

Uses the recipient's public key to protect the message or, in practice, a randomly generated session key

Encrypted Email

Sent through the internet in protected form

Decryption

The recipient uses their private key to recover the session key and decrypt the email

Recipient (Bob)

Reads the original email securely

Steps involved

1. Key generation: The recipient generates an RSA public key and a corresponding private key.
2. Public key sharing: The recipient shares the public key with the sender, preferably through a trusted certificate.
3. Encryption: The sender uses the recipient's public key to encrypt a randomly generated session key. The email content is encrypted using a symmetric algorithm such as AES.
4. Transmission: The encrypted email and encrypted session key are transmitted over the network.
5. Decryption: The recipient uses their RSA private key to recover the session key and then decrypts the email content.

This method is called hybrid encryption. It is efficient because symmetric encryption handles the large email message, while RSA protects the session key.

3. RSA Mathematical Representation

RSA uses the following mathematical operations:

- Encryption: $C=M^e \bmod n$
- Decryption: $M=C^d \bmod n$

Where:

- M = original message or encoded data
- C = encrypted data
- e = public exponent
- d = private exponent
- n = RSA modulus

In practical email systems, RSA is generally used with secure padding and to protect a session key, rather than directly encrypting a complete email.

4. Role of Key Distribution and Management

Key distribution and management are essential for maintaining the confidentiality, authenticity, and integrity of email communication.

1. Secure key distribution: The sender must obtain the correct recipient's public key.
2. Digital certificates: Certificates issued by trusted Certificate Authorities (CAs) help establish the relationship between a user's identity and public key.
3. Private key protection: The private key must be protected using access controls, secure storage, and encryption where appropriate.
4. Key rotation: Keys should be replaced according to the organization's security policy.
5. Revocation: If a private key is compromised, its certificate must be revoked and a replacement key issued.
6. Backup and recovery: Organizations should maintain suitable recovery procedures for encrypted email data without exposing private keys.
7. Digital signatures: RSA signatures can help verify the sender's identity and detect message modification when certificates and keys are properly validated.

Importance of key management

Key management process	Purpose
Key generation	Creates a secure public-private key pair
Distribution	Makes the correct public key available
Authentication	Verifies the identity associated with a key
Secure storage	Prevents unauthorized private key access
Rotation	Limits long-term exposure of keys
Revocation	Invalidates compromised or untrusted keys

Result: RSA-based hybrid encryption, supported by proper key management, provides a practical approach to protecting confidential organizational emails.

5. Email Security Using RSA: Complete Flowchart

1. Recipient generates RSA keys

Public key + Private key

2. Public key is authenticated

Certificate validation or another trusted method

3. Sender prepares the email

Generate a random session key

4. Hybrid encryption

AES encrypts the message; RSA-OAEP protects the session key

5. Send encrypted email

Ciphertext and encrypted session key

6. Recipient decrypts

RSA private key recovers the session key; AES decrypts the message

6. RSA Encryption vs RSA Digital Signature

RSA can support confidentiality and authentication, but these are different operations.

Feature	RSA encryption	RSA digital signature
Main purpose	Confidentiality	Authentication and integrity
Key used by sender	Recipient's public key to protect the session key	Sender's private key to sign
Key used by recipient	Recipient's private key to recover the session key	Sender's public key to verify
Security benefit	Protects message contents	Helps detect modification and verify the signer
Typical padding or scheme	RSA-OAEP	RSA-PSS

A digital signature does not encrypt the email. For strong email security, organizations can combine encryption with digital signatures.

7. Advantages and Limitations of RSA

Advantages	Limitations
Well-established public key algorithm	Requires relatively large keys
Supports encryption and digital signatures	Slower than symmetric encryption
Helps establish secure communication	Private key compromise can expose protected data
Can work with digital certificates	Public keys must be authenticated

Suitable for protecting session keys	Does not provide post-quantum security

Conclusion for Topic 1: RSA helps organizations secure email communication when used with modern padding, hybrid encryption, trusted certificates, digital signatures where needed, and effective key lifecycle management.

The Internet of Things (IoT) consists of connected devices such as smart locks, temperature sensors, security cameras, motion detectors, and smart lights. These devices communicate over networks and often exchange sensitive information.

Cryptography protects IoT data from unauthorized access, modification, impersonation, and other security threats. However, many IoT devices have limited computing power, memory, battery capacity, and communication bandwidth. Therefore, selecting an appropriate cryptographic algorithm is important.

The two approaches considered here are RSA and Elliptic Curve Cryptography (ECC).

9. Constraints of Low-Power IoT Devices



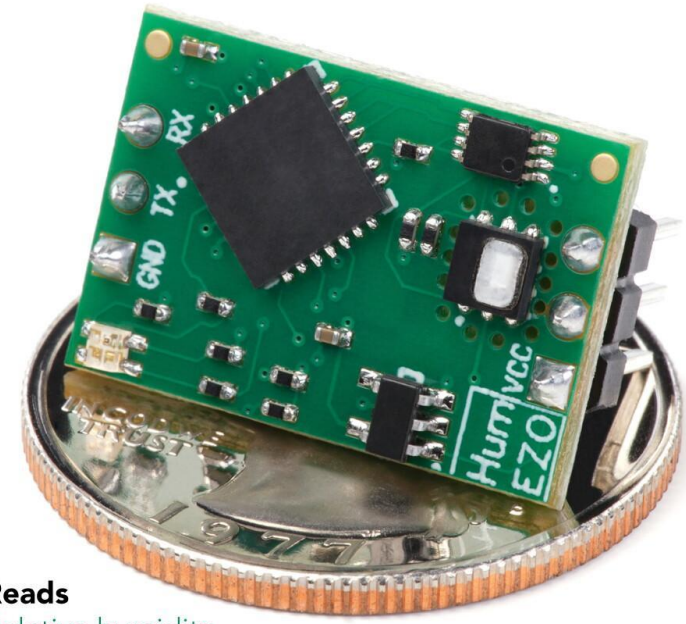
1. Limited battery power

IoT devices may operate on small batteries for months or years. Expensive cryptographic operations can increase energy consumption.



2. Limited processing power

Small microcontrollers may take more time and energy to perform complex mathematical calculations.



Reads
Relative humidity
Dew point
Air temperature

3. Limited memory and storage

Devices may have little RAM and flash storage for cryptographic libraries, certificates, and keys.



4. Network and bandwidth limitations

Wireless devices may use low-bandwidth connections. Large keys and protocol messages increase communication overhead.

Additional challenges include physical tampering, insecure key storage, large numbers of connected devices, and the need for reliable communication in real time.

10. RSA vs ECC: Comparison Table

ECC is a public key cryptography approach based on the mathematical properties of elliptic curves. It can provide comparable classical security to RSA with substantially smaller keys.

Comparison factor	RSA	ECC
Mathematical basis	Integer factorization	Elliptic curve discrete logarithm problem
Typical 128-bit classical security	RSA 3072-bit key	ECC 256-bit key, using an appropriate curve
Key size	Larger	Smaller

Communication overhead	Generally higher	Generally lower
Processing requirements	Can be costly, especially for private-key operations	Often efficient for key agreement and signatures
Memory usage	More storage for larger keys and associated data	Usually lower key-storage overhead
Battery suitability	May be less suitable for constrained devices	Often more suitable
Common applications	Legacy systems, certificates, key transport, signatures	IoT authentication, signatures, and key agreement
Post-quantum security	Not quantum-resistant	Not quantum-resistant

Note: Actual performance depends on the device hardware, implementation, cryptographic scheme, and security parameters. ECC is not automatically faster in every operation.

11. Recommended Approach: ECC

Elliptic Curve Cryptography (ECC) is recommended for low-power smart home IoT devices because it provides strong classical security with smaller public keys and lower communication overhead than RSA at comparable security levels.

For example, a device can use an elliptic-curve key agreement method such as ECDH to establish a shared secret with a gateway. The shared secret is then used to derive symmetric session keys for protecting the actual data.

Reasons for choosing ECC

1. Smaller keys: ECC requires fewer bytes to represent public keys at comparable classical security levels.
2. Lower communication overhead: Smaller keys reduce the amount of data exchanged during connection establishment.
3. Efficient key agreement: ECDH can establish a shared secret without sending the secret itself across the network.
4. Suitable for embedded devices: ECC is widely supported by modern cryptographic libraries and many microcontrollers.
5. Scalable deployment: Smaller messages can help when deploying thousands of connected sensors and smart home devices.

12. IoT Secure Communication Flowchart

Smart Sensor

Generates or securely stores an ECC key pair

Device Authentication

Validates the gateway's identity and credentials

ECC Key Agreement (ECDH)

Both parties derive a shared secret

Session Key Derivation

A KDF derives suitable cryptographic keys

Secure Data Communication

AES-GCM or ChaCha20-Poly1305 protects data confidentiality and integrity

13. RSA vs ECC: Final Decision

Requirement	**Better choice for constrained IoT**
Small public keys	ECC
Low network overhead	ECC
Efficient elliptic-curve key agreement	ECC
Existing infrastructure based on RSA	RSA may be retained where appropriate
Protecting large amounts of sensor data	Symmetric encryption after key establishment
Long-term protection against quantum computers	Neither RSA nor conventional ECC; consider standardized post-quantum cryptography

ECC does not replace every cryptographic component. A complete IoT design should combine ECC-based key establishment with authenticated communication, secure symmetric encryption, device identity verification, key rotation, secure firmware updates, and protected key storage.

14. Conclusion

Public key cryptography is essential for securing email communication and IoT networks. RSA can protect email session keys and support digital signatures, provided that key distribution and management are handled securely. For low-power smart home devices, ECC is generally the better choice because of its smaller keys and lower communication overhead at comparable classical security levels. Combining ECC with efficient symmetric encryption provides a practical balance of security, performance, and energy efficiency.

References

1. Rivest, R. L., Shamir, A., and Adleman, L. (1978). A Method for Obtaining Digital Signatures and Public-Key Cryptosystems.
2. NIST, Recommendation for Key Management: Part 1 – General, SP 800-57 Part 1.
3. NIST, Recommendation for Pair-Wise Key-Establishment Schemes Using Discrete Logarithm Cryptography, SP 800-56A.
4. NIST, Digital Signature Standard (DSS), FIPS 186-5.
5. RFC 8017, PKCS #1: RSA Cryptography Specifications Version 2.2.